Comparison of Global GLP Regulations Pertaining to LC/MS Instrument Software Validation

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Abstract

This paper presents and discusses the global GLP regulations applicable to LC/MS instrument and software used in regulated laboratories. The rationale for this is the globalisation of pharmaceutical research and development as it is important to understand all applicable regulations so that a study can be performed once and used in any regulatory authority worldwide. The main problem with the current GLP regulations is that sophisticated LC/MS systems can be described as apparatus, equipment or computerised system leading to a problem of generating a universal interpretation for qualification and validation of these systems.

Introduction

In today's global business environment, more and more companies are realizing they must focus on multiple marketplaces in order to be successful. Many organisations are also taking advantage of lower costs in emerging markets so that much work that was previously done in the USA or Europe is now being undertaken in Asia, particularly in China and India.

The problem is how to conduct a single activity to meet different regulatory requirements around the world as shown in Figure 1. The key a single approach is an understanding and interpretation of the applicable GXP regulations and guidance coupled with effective risk assessment to enable a unified global approach to an activity. In the case of this paper, it is the topic of computerised system validation for LC/MS instruments and software.



Figure 1: Interpretation of Applicable Regulations Coupled with Risk Assessment Allows a Global Validation Approach

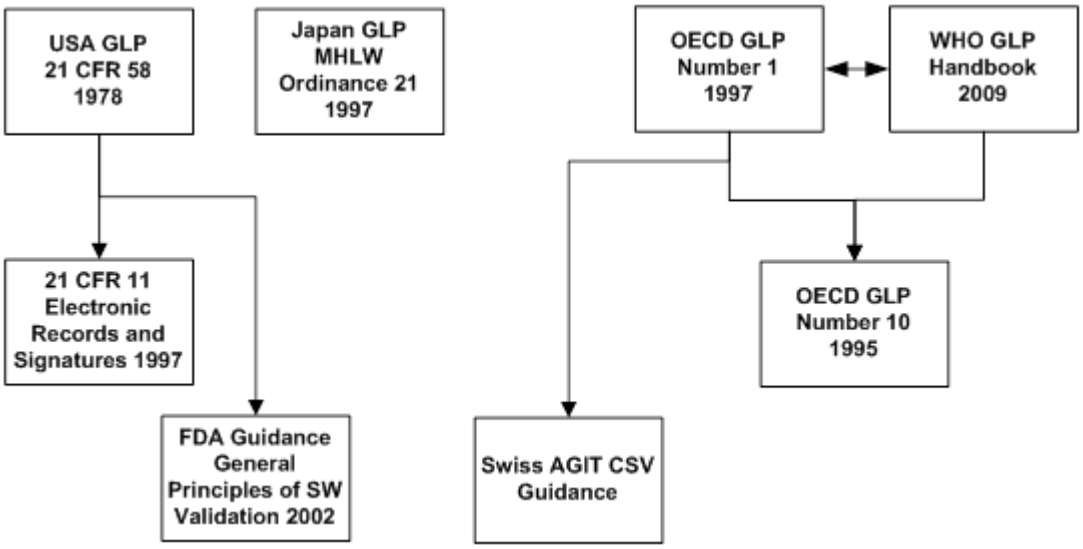
Therefore, for scientific organizations needing to operate in a compliant fashion it is important to understand the similarities and differences between the global regulations, standards, and processes. In this paper we will compare the approaches of four regulations:

- United States Food and Drug Administration (FDA) and the GLP regulations under 21 CFR 58 in combination with 21 CFR 11 regulations for electronic records and electronic signatures
- Japanese GLP issued under Ministry of Heath Labour and Welfare (MHLW) Ordinance 21
- Organisation of Economic Cooperation and Development (OECD) GLP regulations that have been adopted in the majority of OECD member countries with the exception of Japan and USA.
- WHO GLP principles as outlined in their GLP Handbook

In addition, this paper will look at guidance documents issued by the FDA, OECD, and the Swiss AGIT (Working Party for Information Technology) specifically for computerised systems. Therefore the regulatory scope of the paper is shown pictorially in Figure 2.

Then the paper will discuss some of the industry regulatory guidance from the DIA and ISPE to develop a common approach the validation and qualification of LC/MS instruments used in regulated GLP environments.

Figure 2: Scope of the Regulations and Regulatory Guidance covered by this Paper



The goal of this paper is to compare these regulations and guidance documents pertaining to computer system validation (CSV) of software used to operate Liquid Chromatography-Mass Spectrometry (LC/MS) instruments. LC/MS instruments are used extensively for drug discovery and other safety-critical research by pharmaceutical companies, research laboratories, universities, hospitals, and testing facilities around the world. In detailing the requirements from the four regulatory authorities above, a comparison of global regulations can be made. In essence, this is a choice between apparatus, equipment or computerised system which will be discussed in more detail after the presentation of the regulations and regulatory guidance from the four authorities (WHO and OECD regulate but do not inspect).

Definition of Terminology

Four terms need to be defined as they are used within this paper as follows:

- Software Validation: confirmation by examination and provision of objective evidence that software specifications conform to user needs and intended uses, and that the particular requirements implemented through software can be consistently fulfilled [Ref 1].
- Qualification: Action of proving that any equipment works correctly and actually leads to the expected results. [Ref 2]

Validation is focussed on the software and the overall system, whilst qualification is focussed on the instrument and hardware elements of the system and is a subset of the overall validation.

- Software Configuration: changing the business process of the software using tools provided by the supplier, this is GAMP software category 4 [Ref 3]. If a vendor supplies a language for "configuration" of their software it needs to be treated as custom software [Ref 3]
- Software Customisation: this is the writing of modules or software for use with a commercial application or the writing of a custom application. This is GAMP software category 5 [Ref 3].

The LC/MS software discussed in this article is Category 4 or configurable software, no custom software modules are written and all configuration of the application is within the boundaries set by the supplier.

United States Food and Drug Administration (FDA) GLP

Good Laboratory Practice Regulations for Non-Clinical Studies (21 CFR 58)

The Good Laboratory Practice regulations, 21 CFR 58 [Ref 4], have sub part D specifically covering equipment as shown in Table 1. The problem is that the regulations were promulgated in 1978 before widespread introduction and use of computerised systems in laboratories. Therefore the agency and industry has had to interpret computerised system as "equipment" and the regulations have not updated since issue. The three main requirements of this section are that equipment must be of:

- Appropriate design
- Suitably located
- Adequate capacity

These are required so that equipment can do their assigned work, which leaves much room for interpretation of approach. In addition to §58.61, there is also §58.63 covering the requirements for calibration and maintenance of the items of equipment.

Table 1: US GLP Regulations for Equipment [Ref 4]

Clause	Regulation
§58.61	**Equipment design** Equipment used in the generation, measurement, or assessment of data and equipment used for facility environmental control shall be of **appropriate design** and **adequate capacity** to function according to the protocol and shall be **suitability located** for operation, inspection, cleaning and maintenance.

Electronic Signature and Electronic Records final rule (21 CFR 11)

In 1997, the introduction of 21 CFR 11 for electronic records and electronic signatures regulations [Ref 5] allowed companies to work electronically. Part 11 is interpreted by reference to the applicable predicate or existing regulation, in our case 21 CFR 58, therefore if the GLP predicate rule requires a record or signature and a laboratory is working electronically, then the 21 CFR 11 controls are applicable. This requires careful interpretation of the underlying predicate rule for record and signature requirements. In 2003 the FDA issued a guidance for industry entitled Part 11 Scope and Application [Ref 6] that required companies to revisit their interpretation of the regulation and gave enforcement discretion in five areas of the regulation (legacy systems, validation, audit trail, copies of records and record retention). However, if a laboratory wants to work electronically then an audit trail is mandatory, regardless of the recommendations in the FDA guidance document.

Part 11 requires three types of control: administrative, procedural and technical. The first two are the responsibility of the laboratory and the third is the responsibility of the supplier of a computerised system. However, due to the global nature of the pharmaceutical industry if a drug application is submitted to the FDA containing bioanalytical study data performed outside of the USA, the laboratory where the data were generated is subject to Part 11 regulations. Therefore it is important that the software used to generate the bioanalytical data is compliant with the technical requirements of 21 CFR 11.

The Need for 21 CFR Part 11 Assessment of Software

Therefore a 21 CFR Part 11 assessment of the software over the overall system is important and this typically covers technical controls in the application software such as security, access privileges to the application and data, protection of records generated by the system, data integrity and the security and integrity of electronic signatures. In addition the laboratory needs to provide the procedural and administrative controls required by the regulation.

Note, that there are no electronic signature requirements in 21 CFR 11. Signature requirements are found in the underlying predicate rule, 21 CFR 58 [Ref 4]. Where signatures are required 21 CFR 11 defines how an electronic signature must be applied, its meaning and the protection required. Many 21 CFR Part 11 requirements will be met with a combination of technical and procedural controls.

FDA Guidance: General Principles of Software Validation

To overcome the shortcomings and the difficulties in updating their regulations, the FDA issue non-binding recommendations as Guidances for Industry. One of these guidance documents is the General Principles of Software Validation that was issued in final form in 2002 [Ref 1]. The problem is that this was written by the

Center for Devices and Radiological Health (CDRH) where software is not configured, customised or changed as it is part of a regulated medical device. This is different to a bioanalytical laboratory where often software is configured before use to match a laboratory's business process, therefore the whole document, written from the perspective of software development for a medical device must be interpreted with some caution. However, there is some useful guidance for software validation such as:

- Planning is essential for a computerised system validation as it requires a multidisciplinary approach
- User requirements are mandatory to define the intended purpose of the system
- Requirements must be traceable throughout the life cycle
- Testing needs objective documented evidence to support claims that the software meets user requirements

The definition of software validation in this guidance is that provided in the definitions section above, in this it mentions the phrase objective evidence. Objective evidence can be printed provides via screen captures (where they add value), printed reports but this also includes the electronic files generated during the course of the validation.

Japanese Good Laboratory Practice Regulations

The Japanese GLP regulations were issued by under Ordinance 21 by the Ministry of Health, Labour and Welfare [Ref 7] and an unofficial English translation by the Japanese Society for Quality Assurance has been used in preparation of this paper. The applicable regulations for equipment are shown in Table 2. As the US regulations before, only the term equipment is used although in the Japanese regulations there is an implication that computers are involved as there is mention of "collection, measurement and analysis of study records".

Similar to the US GLP regulations there is the requirement for designed appropriately with sufficient processing capabilities. Interestingly there is no requirement for suitable location unlike the US [Ref 4] and the OECD regulations [Ref 8].

Table 2: Japanese GLP Regulations for Equipment [Ref 12]

Section	Regulation
Chapter 3 Testing facilities and equipment (Equipment)	Article 10 Equipment for collection, measurement and analysis of study records, equipment for preserving the environment surrounding the testing facilities, and other equipment necessary to conduct study (hereinafter called the (equipment) shall be **designed appropriately with sufficient processing capabilities**.

OECD Good Laboratory Practice Regulations

The Organisation of Economic Cooperation and Development (OECD) is responsible for developing and maintaining Good Laboratory Practice regulations that are used in the member counties (except USA and Japan) of the OECD for non-clinical, chemical and agrochemical testing studies.

OECD GLP Regulations– 1997

The OECD regulations for LC/MS systems are found in section 4 of the OECD regulations [Ref 8] and shown in Table 3. Instead of equipment the regulations refer to apparatus but reflecting the fact these were written 20 years after the US GLP regulations also include a reference to validated computerised systems. Therefore for the first and only time in a GLP regulation there is a specific requirement for validation of computerised systems. In common with the other GLP regulations apparatus must be:

- Suitably located
- Of appropriate design
- Have adequate capacity

Of interest is the specific requirement that when equipment is calibrated that where possible, standards used are traced to national or international standards of measurement.

Similar to all regulations, there is no mention of how to achieve this as it is left to individual companies to interpret. However the OECD, similar to the FDA produced a guidance document to help companies [Ref 9].

Table 3: OECD GLP regulations for Apparatus [Ref 8]

Section	Regulation
4. Apparatus, Material, and Reagents	1. **Apparatus, including validated computerised systems**, used for the generation, storage and retrieval of data, and for controlling environmental factors relevant to the study should be **suitably located** and of **appropriate design and adequate capacity**.
	2. Apparatus used in a study should be **periodically inspected, cleaned, maintained, and calibrated** according to Standard Operating Procedures. Records of these activities should be maintained. **Calibration** should, where appropriate, be **traceable to national or international standards of measurement**.

OECD GLP Guidance Number 10

The guidance document is a consensus document written by inspectors and representatives of the pharmaceutical industry and issued in 1995. OECD monograph 116 entitled The Application of the Principles

of GLP to Computerised Systems [Ref 9] is a dated document as when it was conceived and written most GLP systems were custom built and few were commercially available. Therefore applying this document has a number of problems, such as:

- No risk assessment principles
- No definition of user requirements
- No mention of software configuration
- Lacks requirements traceability throughout the life cycle

However there are some good principles here such as the operational aspects such as change control, an outline of the SOP required for a system and the elements of a system description. However the latter needs to be interpreted for a commercially available system where much of the required information will not be available for a laboratory.

So where can up to date information be found for risk-based validation of computerised systems used in a GLP environment? Here we need to look at Switzerland.

Swiss AGIT GLP Guidance Documents

The Working Group on Information Technology (AGIT in German) is a collaboration of Swiss inspectors and experts from the pharmaceutical industry working to Good Laboratory Practice (GLP). This group has published four very practical and useful guidance documents on IT within GLP since 2000 [Refs 10 - 14]. The first one on Guideline for the Validation of Computerized Systems [Ref 10] was updated in December 2007 and is an interesting contrast with the Drug Information Association (DIA) Red Apple Guidance [Ref 15] that we will review later in this paper.

Similar to the Drug Information Association (DIA) Red Apple document, there is a high emphasis on the validation plan and report for a system, as the writers equate these documents to a GLP study protocol and report. This is fine, but both publications omit a detailed discussion of the most critical document in the whole validation package: the user requirements specification (URS). If we spent more time and effort on the URS, the corresponding validation efforts and the resulting systems would be so much better and would actually meet business objectives and regulatory requirements.

One area where there are some interesting approaches is the discussion of which systems should be validated, and this is linked to a listing of system categories in Appendix 1. In parallel to USP <1058> on Analytical Instrument Qualification [Ref 16], there are three categories of “system” and, in an attempt to simplify validation; we get some silly issues that arise. For example, for an operating system (OS), no action is recommended and the only documentation is an inventory listing and system description. This is rather short-sighted and contradicts other regulatory expectations from FDA and EMA for qualification of the IT infrastructure [Ref 17].

A company needs to know if the OS of a computer system has been installed correctly, as well as the internet protocol address and other configuration, in order to get the OS to work in a networked environment. In

addition, you also will need to know which service pack has been installed, as this will impact the validation status or the operation of the application that runs on the server or workstation. The same comments apply to databases which are treated the same as an OS. However, both the OS and the database are part of a bigger system and are never used on their own. Installation and qualification of the OS and database should be part of these validations.

In Categories B and C, there are mentions of "restricted customization" and "extended customization" of software, respectively. The problem is that customization is not defined in the guidance document. As defined earlier in this paper, customisation this refers to writing custom code rather than to configuring the software, and it may give readers a different view than the writers anticipated. Perhaps it is better to borrow the definitions from GAMP, and category B should be standard software (either as ROM or installed from CD) and category C should be configured software without custom elements [Ref 3].

Risk management at both the system and functional level is introduced into the validation process. The latter is important, since the authors note that that you may only use 20 percent of the functions of an application. So, why validate the ones you don't use? However, only a single life- cycle model is outlined, which does not give laboratories the full flexibility that the GAMP guide version 5 gives that we will discuss later in this paper. The guidance ignores the need to document the configuration of the software, which is most important in many of the category 4 software applications used in a GLP environment. Also, the vendor audit is placed outside of the usual validation process in a little section at the back of the document. A good feature in Section 9 is an outline of 11 SOPs required to support a computerized system in a GLP environment, although this could be applied to any quality environment [Ref 11].

WHO GLP Handbook second edition 2009

The WHO GLP handbook [Ref 18] is an interpretation of Good Laboratory Practice regulations from the OECD, so much so that 15 OECD publications are published verbatim as Annexes to this document. However as an interpretive document it is useful to see what is required in a little more detail from the OECD GLP regulations. Unfortunately the WHO document does not number their sections in the book which is a really helpful feature. In the sections on Facilities, Buildings and Equipment is a brief section on equipment which is listed in Table 4, this is also complimented by paragraphs on suitability, calibration, maintenance and documentation requirements for equipment. However like all sections equipment has to include computerised systems.

Table 4: WHO GLP Handbook Sections on Equipment and Computerised Systems [Ref 18]

Section	GLP Interpretation
Equipment	For the proper conduct of the study, appropriate **equipment of adequate capacity** must be available. All equipment should be **suitable for its intended use**, and it should be properly calibrated and maintained to ensure reliable and accurate performance. Records of repairs and routine maintenance and of any non-routine work should be retained. Remember that the purpose of these GLP requirements is to ensure the reliability of data generated and to ensure that data are not invalidated or lost as a result of inaccurate, inadequate or faulty equipment.
The Use of Computerised Systems (pp 45-46)	Most laboratories rely heavily on computerised systems for the performance of their studies. Computers are used not only for the collection of data but also for planning and other organisational purposes. The use of computerised systems has undoubtedly increased the efficiency of study performance and the reliability of data. **However, under GLP it is important to demonstrate that such systems are performing correctly, precluding, for instance, the risk of data loss or corruption. This demonstration is called validation** and the OECD has recognised the importance of this work by the publication of a specific guideline on this subject (The Application of The Principles of GLP to Computerised Systems) appended to this Handbook and discussed in the training manuals.

On pages 45 and 46 of the WHO handbook there is a specific reference to the use of computerised systems in GLP studies unfortunately this cross refers to the outdated OECD monograph for computerised systems [Ref 9].

GLP Regulatory Summary: Apparatus, Equipment or Computerised System?

The GLP regulations issued between 1978 and 1997 are showing their age as they have not moved from the initial approach of the US GLP regulations [Ref 4] that define computerised systems as equipment. References within the GLP regulations are to either apparatus or equipment: mixing computerised systems with pipettes and glassware. There are a few references to computerised systems as can be seen from Table 5.

Table 5: Comparison of Terms for the Major GLP Regulations

Agency	FDA [Ref 4]	OECD [Ref 8]	Japan [Ref 7]	WHO [Ref 18]
Equipment	✓		✓	✓
Apparatus		✓		
Computerised System		✓		✓

Where computerised systems have guidance documents available they need to be interpreted with care [Ref 2] or are outdated [Ref 9]. Therefore to find better information on qualification of the instrument and validation of the software we need to look outside of the regulatory authorities and to industry guidance.

Industry Guidance for Computerised System Validation

Red Apple 2 Guidance Document

One of the oldest computerised system validation guidance documents is entitled Computerised Systems in Used in Non-Clinical Safety Assessment [Ref 19], otherwise known as the Red Apple Document. This was originally published in 1988 so it is out of date, however following a second Red Apple conference in 2006, a new version of the guidance was published in May 2008 [Ref 15]. As with the original publication it is intended for facilities working to Good Laboratory Practice (GLP) and is published by the Drug Information Association (DIA).

The updated publication is divided into eight chapters [Ref 19]:

- System Life Cycle
- Validation Plan
- Quality in a Regulated Environment
- Security
- Long Term Retention of Electronic Records
- Business Continuity and Disaster Recovery
- Infrastructure
- Instruments and Computerised Equipment

The first three chapters discuss aspects of computer validation, the next three chapters the protection of the records generated by a system and the last two chapters important concepts for the application of IT and instruments in a GLP environment. The arrangement of the chapters appears haphazard rather than planned in an organised sequence. The main problem with this approach is the major focus of the publication on the validation plan (equivalent to a protocol) to the exclusion of effective risk management, flexible system life

cycle models and failing to document user requirements adequately. There is also no mention of traceability of user requirements throughout the rest of the life cycle. As such the whole publication misses the point and provides the wrong guidance to a GLP laboratory.

To give a specific example of the many flaws in this document is to be found in Chapter 2 on validation plans. The table entitled matrix 4 shows the tasks and responsibilities for the installation and qualification of a firewall in an IT infrastructure. There are up to 24 separate qualification tasks including writing SOPs, but this is for a firewall, a standard item of IT hardware and software that should be qualified as equipment, which is a simpler process, and not validated as if it were software. If everyone did this, then regulated companies would be out of business. There is no effective risk management just to generate a pile of documents for a firewall. It is approaches like this that give computerised system validation a bad name and fail to deliver business benefit for an organisation.

GAMP® 5 Risk-Based Validation Guidance

GAMP stands for Good Automated Manufacturing Practice. The original concept, originated by the UK pharmaceutical industry twenty years ago, was to control suppliers of machines and systems for pharmaceutical manufacture i.e. GMP facilities. However once it was adopted by the International Society of Pharmaceutical Engineering (ISPE) is has moved out of the UK and is now a global approach to validation. The current version of the GAMP guide, version 5, was published in 2008 [Ref 3] and contains a risk based approach to computerised system validation. This is evidenced in the different life cycle models for different categories of software (category 3, category 4, Category 4 with category 5 custom modules and category 5 applications) with increasing documentation requirements with increasing software risk. Also there is a section on leveraging vendor testing to focus user testing on critical functions for an individual laboratory.

The most important update in GAMP 5 [Ref 3] is the focus on managing the risk posed with a computerised system and the impact the system can have on patient safety, product quality, or data integrity [Ref 1]. The latter is important as GLP is often seen as far away from product quality and patient safety but data integrity is important especially considering the fraud and falsification issues with Able Laboratories [Ref 20, 21] and Cetero [Ref 22] just to name two that have been found between 2005 and today. Hence the FDA has trained their inspectors to look at computerised systems in more detail and not just to focus on paper records. Therefore in a GLP environment the focus of the validation should be on demonstrating intended purpose and ensuring data integrity.

GAMP Good Practice Guide for Computerised Systems, Second Edition

Recently, the second edition of the GAMP Good Practice Guide (GPG) for Risk Based validation of Computerised Laboratory Systems [Ref 23] has recently been published by ISPE. While the main GAMP 5 guide is general, the GPG focuses specifically on laboratory computerised systems and outlines in appendices how to apply the principles in the main guide to laboratory computerised systems of varying complexity. For complex systems, the system illustrated in automated dissolution testing and HPLC analysis controlled by a chromatography data system (CDS). In this appendix there is a requirement for documenting the configuration of the CDS software [Ref 23]

Importance of Risk Assessment

Both GAMP 5 [Ref 3] and the Laboratory GPG [Ref 23] use a risk assessment methodology called qualitative Failure Mode Effect Analysis (FMEA) which is better suited to category 5 rather than category 4 software. The methodology assesses the probability and severity of harm if a fault occurs to classify and rank risks, rather than the probability of the fault occurring. For each identified risk the probability of detection by the system is estimated and plotted to find the most important risks to mitigate (typically high risk ranking with low probability of detection by the system. After a specific risk is mitigated, the risk assessment is repeated to assess if the mitigation is successful. This allows the laboratory to focus their validation effort on specific high risk areas of the system according to their intended use requirements.

If the risk assessment indicates that software controlling the data acquisition contains more risk than application functions used to create a template for reporting the results of an analytical run, the controls and risk mitigation will be focussed at data acquisition rather than the report template. However in a commercial LC-MS system, a laboratory may decide to leverage the work performed by the vendor during development and at installation qualification to reduce testing here and focus validation resources on the report template instead. All that is required is a justified and documented risk assessment.

A Global Validation Approach to Meet Global Regulations

General Considerations

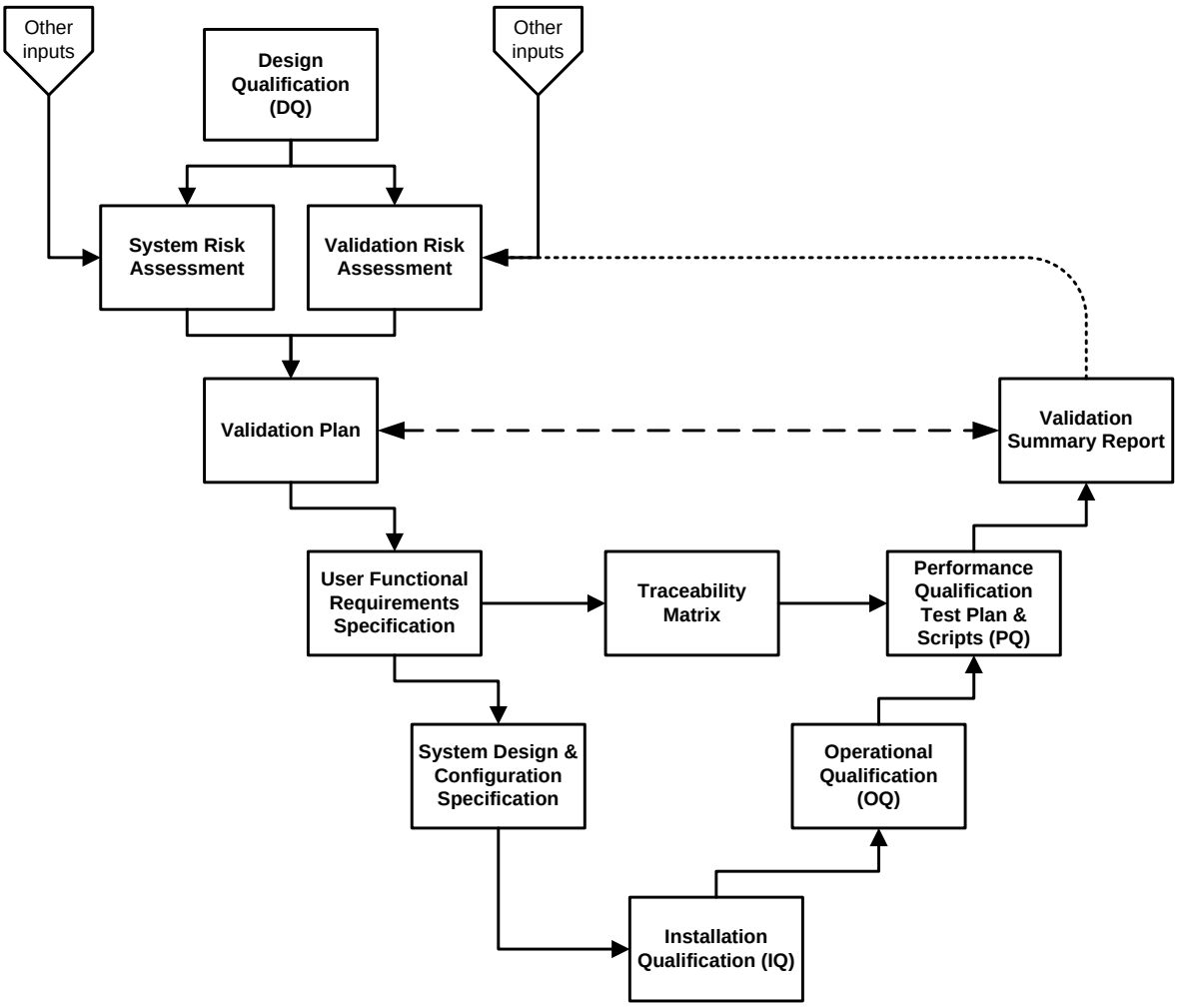
To ensure that a single validation meets global GLP regulations and good practice requirements a number of factors need to be considered such as:

- Planning and reporting the validation effort
- Specifying user requirements and documenting the system configuration for intended use
- Qualifying the supplier
- Ensuring traceability of requirements throughout the life cycle
- Installing, integrating and qualifying the components to form a system
- Testing the configured system versus intended use requirements
- Writing procedures and training users
- Releasing the system for operational use

A System Life Cycle to Meet Global GLP Requirements

The overall system life cycle used for a global validation is shown in Figure 3 and is discussed below.

Figure 3: System Life Cycle and Documented Evidence for a Global Approach to LC/MS Validation



System Validation Documents

The following list of documents reflects a typical software validation document set that would meet global GLP requirements.

- Validation Risk Assessment (VRA) – a risk assessment based on GAMP 5.0 that determines if the system needs to be validated or not.
- Validation Plan (VP) – comprehensive plan for conducting validation that defines the scope of the validation effort, the roles and responsibilities of who is involved in the work and the life cycle to be followed with the associated documented evidence to be produced.
- User Functional Requirements Specification (UFRS) – list of user and functional requirements to meet design specifications. Particular user requirements are typically devised via a GAMP 5 risk-based

approach and they are documented in the User and Functional Requirement Specification (UFRS). This document defines the intended use of the system.

- System Design & Configuration Specification (SDCS) – description or "blueprint" of the system
- Test Plan (TP) – list of tests to be conducted during validation
- Design Qualification (DQ) Protocol – verification that software has been designed to meet intended use
- Installation Qualification (IQ) Protocol – test scripts to verify that software has been configured to meet design specifications
- Operational Qualification (OQ) Protocol – test scripts to verify that software operates in accordance with design specifications
- Performance Qualification (PQ) Protocol – test scripts to verify that software consistently meets performance specifications when operating under normal conditions
- Traceability Matrix (TM) – method of tracking the user and functional requirements to the test scripts
- Validation Summary Report (VSR) – comprehensive summary of validation activities and the results of same

These validation deliverables are sufficient to meet the majority of global regulatory requirements and validation best practices.

Conclusions

With rampant globalization, similarities in the different global regulations that govern LC/MS software validations are becoming frequently encountered. Though the remaining differences and evolving changes to the regulations are forcing the laboratory and system vendor to focus on a "superset" of regulations to monitor their risk, they must also adhere to local/required regulations and requirements as the foundation for the validation plan and execution of the validation project.

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